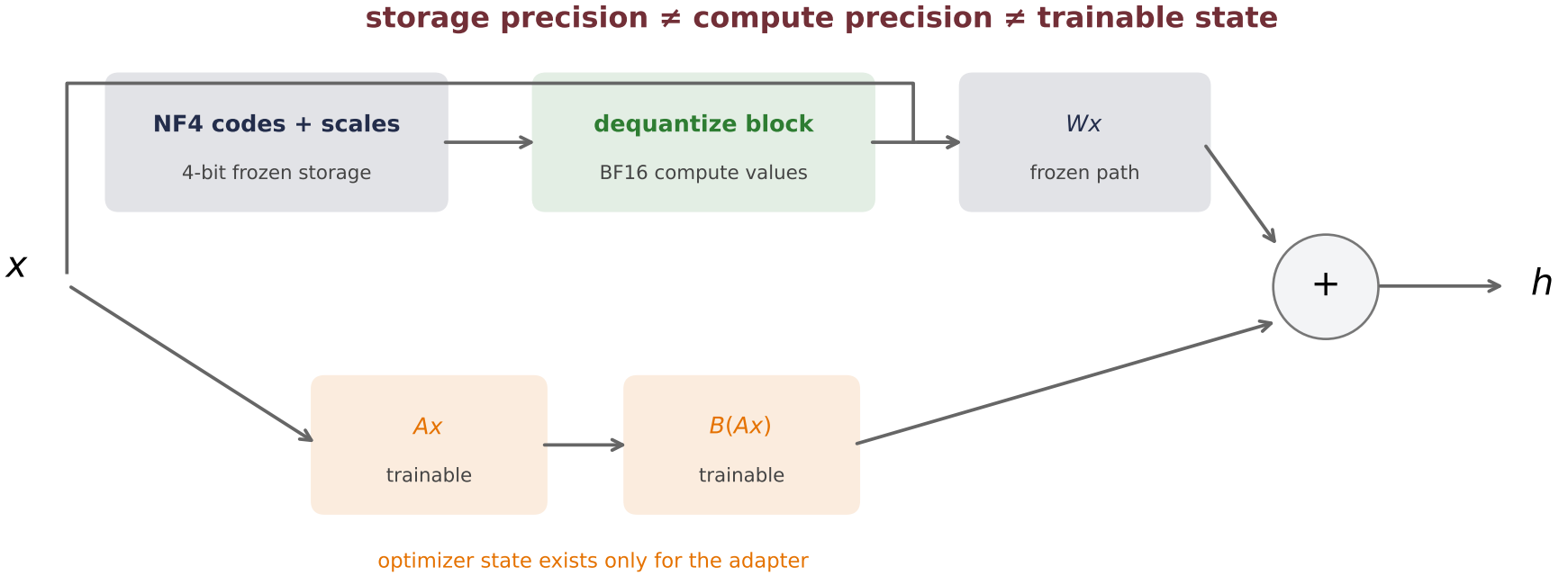
storage precision ≠ compute precision ≠ trainable state
NF4 codes + scales
4-bit frozen storage
dequantize block
BF16 compute values
Wx
frozen path
x
+
h
Ax
trainable
B(Ax)
trainable
optimizer state exists only for the adapter